// EFFAT UNIVERSITY · COLLEGE OF ENGINEERING

CS3081 · ARTIFICIAL INTELLIGENCE

RESOLUTION HACKATHON

> case_file: the_missing_exam_paper

In-class team sprint · 50 min · 4 levels · 100 XP + bonus

Mentor: Dr. Naila Marir

incident_report.txt

```
$ cat incident_report.txt

Monday 08:00. The CS3081 final exam paper is gone from the instructor's office.
Three students were seen near the office on Sunday night:
  → Sara   → Omar   → Lina

Campus security left 7 clues. Your team has 50 minutes to hack the case with
resolution:
find who took it, who is innocent, and which accusation cannot be proven.

$ ./start_hackathon --teams 3-4 --mode proof_by_contradiction
```

// TEAM REGISTRATION

TEAM NAME		TABLE #	
HACKER 1	Busra Suleymanoglu	HACKER 2	
HACKER 3		HACKER 4	

// THE 4 LEVELS

LEVEL 01
CNF Relay
10 min · 20 XP

LEVEL 02
Resolution Dominoes
10 min · 30 XP

LEVEL 03
Solve the Case
20 min · 40 XP

LEVEL 04
Final Accusation
10 min · 10 XP

Bonus XP: +3 FIRST BLOOD per level · +3 BUG HUNTER per accepted objection · +3 SIDE QUEST in Level 03

README.md

Rules of the Hack

50 MIN | 100 XP

▶ Pick your roles — rotate every level

C
CONVERTER
Turns logic into CNF: eliminate ↔ and →, De Morgan, distribution.

R
RESOLVER
Hunts complementary literals and builds new clauses (resolvents).

✓
CHECKER
Debugs every step and writes the final answer. Nothing ships without the Checker.

▶ How the game runs

01 Teams of 3–4. Play the four levels in order.

02 When your team clears a level, raise your hand. The mentor checks it and unlocks the next level.

03 **First Blood:** the first team to clear each level correctly gets **+3 XP**.

04 Allowed moves only: eliminate ↔, eliminate →, De Morgan, distribution, resolution and factoring.

05 Show every step. A correct answer with no steps earns only half the XP.

▶ XP table

LEVEL	CHALLENGE	TIME	XP
01	CNF Relay — convert the clues	10 min	20
02	Resolution Dominoes — match clause cards	10 min	30
03	Solve the Case — proof by contradiction	20 min	40
04	Final Accusation — present and object	10 min	10 + steals
	Total	**50 min**	**100 + bonus**

DATASET

The Evidence Board

10 SYMBOLS | 7 CLUES

▶ Symbol table

SYMBOL	MEANING
D	The office door was unlocked
S	Sara took the exam paper
O	Omar took the exam paper
L	Lina took the exam paper
C	The security camera was switched off
F	The office door was forced open
Y	Lina had a copy of the office key
K	Khalid lent his office key
A	The alarm rang

▶ Clues from campus security

#	CLUE (ENGLISH)	LOGIC
1	If the door was unlocked, then Sara or Omar took the paper.	$D \rightarrow (S \vee O)$
2	The door was unlocked.	D
3	If Sara took the paper or the door was forced, then the camera was off.	$(S \vee F) \rightarrow C$
4	The camera was NOT off (it recorded all night).	$\neg C$
5	Lina could only have taken it if she had a copy of the key.	$L \rightarrow Y$
6	Lina had a copy of the key if and only if Khalid lent his key.	$Y \leftrightarrow K$
7	It is not true that Khalid lent his key AND the alarm rang.	$\neg(K \wedge A)$

CHEAT CODE

To show KB ⊨ α: add **¬α** to the KB, convert everything to CNF, and keep resolving. If you produce the empty clause **( )**, it's a contradiction, so KB ⊨ α. If no new clauses can be made and there is no ( ), there is **no entailment**.

LEVEL 01 / 04

CNF Relay

10 MIN 20 XP

RELAY Pass the sheet around the team. **Each hacker writes ONE step, then passes it on.** Write the rule you used next to each step.

▶ Clue 1: D → (S ∨ O) [4 XP]

STEP	EXPRESSION	RULE USED	WRITTEN BY
1	~D v (S v O)	Eliminate	
2	~D V S V O		
3	clause (~D V S V O)	CNF	

▶ Clue 3: (S ∨ F) → C [6 XP]

STEP	EXPRESSION	RULE USED	WRITTEN BY
1	~(S v F) v C	Eliminate Implication	
2	(~S ∧ ~F) v C	De Morgan's law	
3	(~S v C) ∧ (~F v C)	Distribution	

▶ Clue 6: Y ↔ K [6 XP]

STEP	EXPRESSION	RULE USED	WRITTEN BY
1	(Y → K) ∧ (K → Y)	Eliminate Biconditional	
2	(~Y v K) ∧ (~K v Y)	Eliminate Implications	
3	(~Y v K) ∧ (~K v Y)	CNF	

▶ Clue 7: ¬(K ∧ A) [4 XP]

STEP	EXPRESSION	RULE USED	WRITTEN BY
1	~K v ~A	De Morgan's Law	
2	(~K v ~A)	final CNF	
3			

▶ Compile the final KB in CNF

List every clause from all seven clues, one per row, as K1, K2, K3, … You need these numbers in Level 03.

NO.	CLAUSE		NO.	CLAUSE
K1	~D v S v O	clue 1	K6	~L v Y
K2	D	clue 2	K7	~Y v K
K3	~S v C	clue 3	K8	~K v Y
K4	~F v C	clue 5	K9	~K v ~A
K5	~C	clue 4	K10	only 9 clauses

LEVEL 02 / 04

Resolution Dominoes

10 MIN 30 XP

Cut out the 12 clause cards on the next page and shuffle them. Then:

01 Find pairs of cards that share a **complementary literal** (like P and ¬P). Each card belongs to exactly one pair.

02 Place the two cards side by side like dominoes and write the resolvent below.

03 If the resolvent has a duplicate literal, apply **factoring** and say so.

⚠ BUG One pair is a **trap**. It looks like it gives ( ) but it doesn't. Think carefully about what one resolution step is allowed to do.

PAIR	CARD + CARD	RESOLVED ON	RESOLVENT	FACTORING?
1	B + G	T	(U ∨ U) → U	(Y) / N
2	C + H	Q	P ∨ R	Y / (N)
3	D + J	Y	(T ∨ Z ∨ W)	Y / (N)
4	E + L	M	() empty clause	Y / (N)
5	F + K	A	(B ∨ S ∨ R ∨ S) → B ∨ R ∨ S	(Y) / N
6	A + I A Trap?	G (On H)	(H ∨ ¬H) → tautology	Y / (N)

▶ Bug report: the trap pair

Which pair is it? Explain in one or two sentences why it does NOT give the empty clause.

Pair 6: The trap pair is Card A and Card I. Resolving them on G gives H ∨ ¬H, not the empty clause. One resolution step can remove only one complementary pair, so we cannot cancel both G and H at the same time.

LEVEL 02 · ASSETS

Clause Cards

CUT ON DASHED LINES

CARD_A (G ∨ H) { resolve me }	CARD_B (¬T ∨ U) { resolve me }	CARD_C (¬Q ∨ R) { resolve me }
CARD_D (X ∨ ¬Y ∨ Z) { resolve me }	CARD_E (M) { resolve me }	CARD_F (A ∨ B ∨ S) { resolve me }
CARD_G (T ∨ U) { resolve me }	CARD_H (P ∨ Q) { resolve me }	CARD_I (¬G ∨ ¬H) { resolve me }
CARD_J (Y ∨ W) { resolve me }	CARD_K (¬A ∨ R ∨ S) { resolve me }	CARD_L (¬M) { resolve me }

LEVEL 03 / 04

Solve the Case

20 MIN 40 XP

Use your CNF KB (K1, K2, ...) from Level 01. For each case, add the **negation** of what you want to prove, then resolve. Number new clauses R1, R2, R3, ...

▶ Case 3A — Did Omar take the paper? Prove KB ⊨ O [15 XP]

ADD TO KB:

STEP	CLAUSE 1 (NO.)	CLAUSE 2 (NO.)	RESOLVED ON	NEW CLAUSE (RESOLVENT)
R1	K1	K2	D	(S ∨ O)
R2	R1	K10	O	(S)
R3	K3	K5	C	(¬S)
R4	R2	R3	S	□
R5				

VERDICT:

▶ Case 3B — Is Sara innocent? Prove KB ⊨ ¬S [10 XP]

ADD TO KB:

STEP	CLAUSE 1 (NO.)	CLAUSE 2 (NO.)	RESOLVED ON	NEW CLAUSE (RESOLVENT)
R1	K10	K3	S	(□)
R2	R1	K5	C	□
R3				

VERDICT: empty clause → contradiction → KB ⊨ ¬S
Sara is innocent

LEVEL 03 / 04 · BOSS FIGHT

Case 3C — What about Lina?

15 XP

interrogation.log

```
Omar: "Lina did it with me!"
Lina: "I am innocent!"
$ try prove(L) && try prove(¬L)
```

▶ Attempt 1 — add ¬L and resolve

STEP	CLAUSE 1 (NO.)	CLAUSE 2 (NO.)	RESOLVED ON	NEW CLAUSE (RESOLVENT)
R1	K_6	K_7	Y	$(\neg L \vee K)$
R2	R_1	K_9	K	$(\neg L \vee \neg A)$
R3	K_1	K_2	D	$(S \vee O)$
R4	K_2	K_5	C	$(\neg S)$

▶ Attempt 2 — add L and resolve

STEP	CLAUSE 1 (NO.)	CLAUSE 2 (NO.)	RESOLVED ON	NEW CLAUSE (RESOLVENT)
R1	K_{10}	K_6	L	(Y)
R2	R_1	K_7	Y	(K)
R3	R_2	K_9	K	$(\neg A)$
R4	R_L	K_8	K	(Y)

▶ Conclusion

Can the detectives accuse Lina? Can they clear her? What does this mean about entailment?
Hint: look at where the literal L appears in the KB.

The detective cannot prove that Lina took the paper or that she is innocent - Adding either L or ¬L doesn't produce the empty clause. Therefore, the KB entails neither L nor ¬L, meaning there is not enough evidence to decide whether Lina took the paper.

SIDE QUEST — Finished early? Prove that the door was **not** forced: KB ⊨ ¬F. Write your proof on the back of this page for **+3 XP**.

Side Quest: The door was not forced open because assuming F leads to a contradiction. Resolving F with $\neg F \vee C$ gives C, and resolving C with $\neg C$ gives the empty clause. Therefore, KB ⊨ ¬F

LEVEL 04 / 04

Final Accusation

10 MIN 10 XP

01	The mentor picks one case (3A, 3B or 3C) for your team. Present it on the board in **2 minutes**.
02	While a team presents, the other teams are **code reviewers**. They look for an invalid step.
03	To object, hold up your **OBJECTION!** card and name the exact step and the broken rule.
04	A **correct objection steals 3 XP** from the presenting team. A wrong objection costs 2 XP.
05	A clean presentation with no accepted objection earns the full 10 XP.

▶ Code review log

AGAINST TEAM	STEP	WHAT WAS WRONG	ACCEPTED?

▶ Post-mortem (individual, one sentence each)

Why is resolution called a “proof by contradiction”? What does it mean when no empty clause can ever be produced?

To check $KB \models \alpha$ we add $\neg\alpha$, producing $\square$ (equivalent to False) show $KB \wedge \neg\alpha$ is a contradiction, so $KB \models \alpha$. If we can't add any new clauses and never produce $\square$, $KB \wedge \neg\alpha$ is not a contradiction, so there is no entailment: $KB \not\models \alpha$

UNLOCKABLES

Achievements & Leaderboard

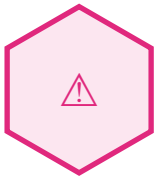
TICK WHAT YOU EARN



FIRST BLOOD

First team to clear a level. +3 XP each time.

□ earned



TRAP DETECTOR

Spotted and explained the trap pair in Level 02.

□ earned



SIDE QUEST

Proved KB ⊨ ¬F in Level 03. +3 XP.

□ earned



CLEAN BUILD

Presented in Level 04 with zero accepted objections.

□ earned

▶ Team scoreboard (mentor use)

L01	L02	L03	L04	BONUS	STEALS	TOTAL
/20	/30	/40	/10	+	±	**/100**

// cut out and hold up during Level 04

ERROR 403: INVALID STEP

OBJECTION!

step ____ breaks rule: _____